

Question 2 : Weighted Summing Amplifier using Operational Amplifier

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Introduction

0.1 Objective

Our goal is to design a system using single Operational Amplifier (Op-Amp) whose output voltage that produces

$$V_{out} = 2V_1 - V_2$$

where V_{out} is the output voltage and V_1 , V_2 are input voltages

We will use **NGSPICE** to run this simulation. For the final submission, the input voltages will be defined as $V_1 = 3u(t)$, and $V_2 = 2u(t)$, where $u(t)$ represents a unit step signal. Next, we will perform a transient analysis and generate a plot showing how the output voltage V_{out} , changes over time.

0.2 Circuit

We will use the following circuit diagram and compute the relationship between resistors to obtain an output voltage .

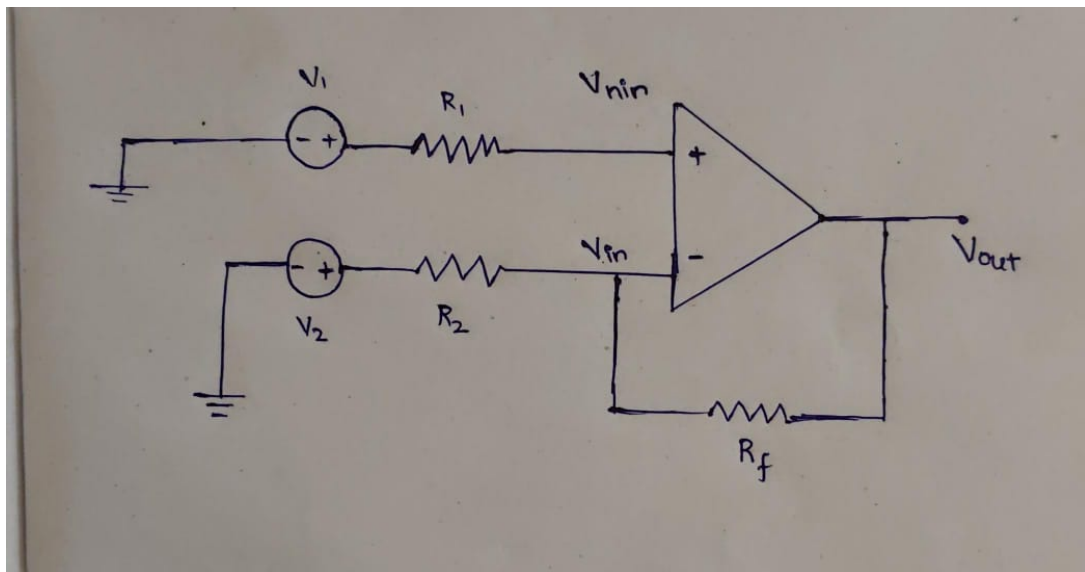


Figure 1: Circuit Diagram

Theoretical Derivation

We obtain the expression for V_{out} using **Virtual Short**
From Virtual Short principle ,

$$V_{in} = V_{nin}$$

where V_{in} is the voltage at the inverting terminal and V_{nin} is the voltage at the non-inverting terminal.

If the non-inverting input is connected to a voltage source V_1 (No current enters the input, so the resistor R_1 carries no current. Thus ,the node voltage at non-inverting input equals the source) :

$$V_{in} = V_{nin} = V_1 \quad (1)$$

Applying **KCL** at the inverting node ,

$$\frac{V_{in} - V_2}{R_2} + \frac{V_{in} - V_{out}}{R_f} = 0$$

$$\frac{V_1 - V_2}{R_2} + \frac{V_1 - V_{out}}{R_f} = 0 \quad (\text{from (1)})$$

$$V_{out} - V_1 = \frac{-R_f}{R_2}(V_2 - V_1)$$

$$V_{out} = V_1 + \frac{R_f}{R_2}V_1 - \frac{R_f}{R_2}V_2$$

$$V_{out} = (1 + \frac{R_f}{R_2})V_1 - (\frac{R_f}{R_2})V_2$$

To get our objective , we would have require the feedback resistance R_f equal to R_2

The Code

```
*NGSPICE Question 2
*Weighted Summing Amplifier using Operational Amplifier

*loads the model file Lm358.mod for simulation
.include "Lm358.mod"

*providing the positive and negative power supply for Op-Amp
VCC Vcc 0 15
VEE Vee 0 -15

*Resistor components
R1 V_1 V_nin 10k
R2 V_2 V_in 50k
Rf V_in V_out 50k

*defining input voltages using PULSE
*PULSE(intial_voltage pulse_voltage delay_time rise_time fall_time pulse_width
period) - format
V1 V_1 0 PULSE(0 3 0 1ns 1ns 100 2)
V2 V_2 0 PULSE(0 2 0 1ns 1ns 100 2)

* Op-Amp Subcircuit for intialising Lm358 model
* order : (+) (-) VCC VEE out model_name
XOPAMP V_nin V_in Vcc Vee V_out LM358/NS

*control block
.control
tran 0.001ms 8ms ; transient analysis
plot v(V_1) v(V_2) v(V_out) ; plotting the voltages
.endc

.end
```

Code Analysis

A detailed explanation of the SPICE code for simulating a weighted summing amplifier using the LM358 model file , implementing the equation

$$V_{out} = 2V_1 - V_2$$

0.2.1 Title and Including Model

```
*NGSPICE Question 2
*Weighted Summing Amplifier using Operational Amplifier
*loads the model file Lm358.mod for simulation
.include "Lm358.mod"
```

Explanation :

- ◇ In **NGSPICE**, lines starting with * are comments which are not considered during execution.
- ◇ The command `.include "LM358.mod"` imports the model file which contains the detailed behaviour and characteristics of the OP-Amp.

0.2.2 Power Supply

```
*providing the positive and negative power supply for Op-Amp
VCC Vcc 0 15
VEE Vee 0 -15
```

Explanation :

- ◇ These are independent DC sources. V_{CC} makes node $V_{cc} = +15$ V relative to ground 0. V_{EE} makes node $V_{ee} = -15$ V.
- ◇ The OP-Amp will use those nodes as positive and negative supply pins.

0.2.3 Resistor Network

```
*Resistor components
R1 V_1 V_nin 10k
R2 V_2 V_in 50k
Rf V_in V_out 50k
```

Explanation :

- ◇ R1 connects the top input node V_1 to the non-inverting input node V_{nin} .
- ◇ R2 connects V_2 to the inverting input node V_{in} .
- ◇ Rf is the feedback resistor from output V_{out} back to the inverting node V_{in} .

0.2.4 Input Sources

```
*defining input voltages using PULSE
*PULSE(intial_voltage pulse_voltage delay_time rise_time fall_time pulse_width
period) - format
V1 V_1 0 PULSE(0 3 0 1ns 1ns 100 2)
V2 V_2 0 PULSE(0 2 0 1ns 1ns 100 2)
```

Explanation :

- ◇ Both voltage sources generate pulse waveforms using the PULSE function.
- ◇ **For V_1 :** starts at 0 V, pulses to 3 V, no initial delay, rise/fall = 1 ns, ON time = 100 (units depend on time unit you use later), period = 2.
- ◇ **For V_2 :** same shape but pulses to 2 V.

0.2.5 Op-Amp subcircuit instantiation

```
* Op-Amp Subcircuit for intialising Lm358 model
* order : (+) (-) VCC VEE out model_name
XOPAMP V_nin V_in Vcc Vee V_out LM358/NS
```

Explanation :

- ◇ XOPAMP is the subcircuit name
- ◇ The line indicates :

$V_{nin} \rightarrow$ non - inverting input

$V_{in} \rightarrow$ inverting input

$V_{CC} \rightarrow$ positive supply

$V_{EE} \rightarrow$ negative supply

$V_{out} \rightarrow$ output node

- ◇ LM358/NS references the specific op-amp model from the included file.

0.2.6 Control Block Simulation commands

```
*control block
.control
tran 0.001ms 8ms ; transient analysis
plot v(V_1) v(V_2) v(V_out) ; plotting the voltages
.endc

.end
```

Explanation :

- ◇ `.control` / `.endc` encloses simulator commands
- ◇ `tran 0.001ms 8ms` simulates transient analysis from 0 to 8 ms with a maximum step of 0.001ms - it computes time-domain response.
- ◇ `plot v(V_1) v(V_2) v(V_out)` draws the three node voltages so you can visually compare inputs and output.
- ◇ `.end` marks the end of the netlist.

Simulation Plot

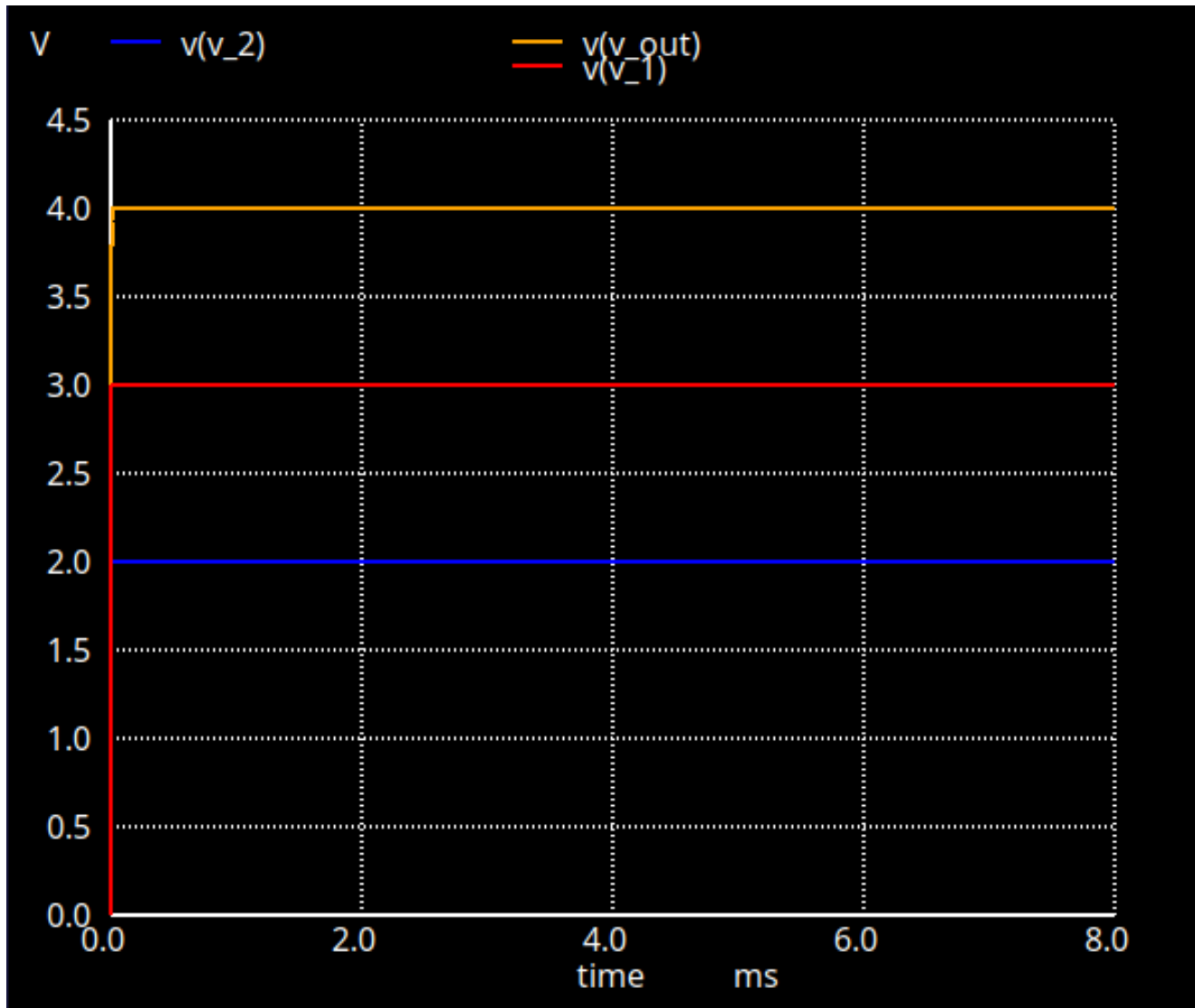


Figure 2: transient response plot

Circuit Analysis

Theoretical Output

For the circuit , the output voltage

$$V_{out} = (1 + \frac{R_f}{R_2})V_1 - (\frac{R_f}{R_2})V_2$$

With $R_1 = 10\text{ k}\Omega$, $R_2 = R_f = 50\text{ k}\Omega$.

$$V_{out} = (1 + \frac{50k}{50k})V_1 - (\frac{50k}{50k})V_2$$

$$V_{out} = 2V_1 - V_2$$

Substituting , $V_1 = 3V$ and $V_2 = 2V$

$$V_{out} = (2)(3) - (1)(2)$$

$$V_{out} = 4V$$

Transient Response

The transient analysis helps us understand how the weighted summing amplifier behaves when its input voltages change suddenly. In this circuit, the two inputs V_1 and V_2 are square-wave signals that switch between 0–3 V and 0–2 V. These step changes allow us to observe how quickly and accurately the op-amp responds under closed-loop operation.

- When the inputs transition to their high levels, the op-amp output responds according to the closed-loop expression

$$V_{out} = 2V_1 - V_2.$$

For $V_1 = 3\text{ V}$ and $V_2 = 2\text{ V}$, the output settles at 4 V. When the inputs return to zero, the output also returns to 0 V. The waveform therefore forms a clean 0–4 V square wave.

- The simulated waveform shows that the output reaches its final value almost instantly after each input transition.

- The LM358's built-in frequency compensation, combined with the feedback configuration, keeps the amplifier well-behaved during transitions. This results in smooth, stable movement between the high and low output levels.
- During the high part of the input pulse, the amplifier holds the output steady at $4V$. When the inputs are low, the output remains at $0V$.

Conclusion

The transient results match the analytical expression $V_{out} = 2V_1 - V_2$. The amplifier transitions cleanly between levels, showing stable and predictable behavior.